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Mechanistic modelling of gastric evacuation in predatory gadoids applying the square root model to describe surface-dependent evacuation

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A new model approach to gastric evacuation in predatory fishes was shown to give accurate and reliable estimates of evacuation. It is believed that such a model would prove particularly useful in understanding predator-prey interactions in natural systems. The model is a simple, geometric abstraction of the square root model predicting that evacuation time is proportional to the square root of meal size in accordance with extensive empirical evidence. Digestive processes are assumed to be restricted to an outer surface of total stomach content that is represented by the curved side of a cylinder. This way, total stomach content is considered a cylinder of constant length the mass of which is gradually reduced by successive peeling off its side leaving the ends unaffected. The fundamental rate δ describes mean thickness of the layer that is peeled off the cylinder per time unit. In the model, δ is constant independent of cylinder thickness but inversely proportional to the square root of cylinder length. Anatomical and dynamic characteristics of the stomach render this formulation of δ plausible. Using mean evacuation rate over time during evacuation of a meal, the model disregards prey heterogeneities that influence instantaneous evacuation rate. Relationships between prey heterogeneities and evacuation patterns were indicated by application of a general power model to evacuation data, and it was suggested how the effects of prey characteristics might be incorporated into the square root model to produce a generic model of gastric evacuation. © 2005 The Fisheries Society of the British Isles

Key words: cylindrical prey geometry; gastric evacuation model; prey heterogeneity; prey homogeneity; surface-dependent process.

INTRODUCTION

This study aimed at providing a mechanistic model approach to gastric evacuation that gives accurate and reliable estimates of evacuation in wild predatory fishes. Such a model should prove particularly useful in understanding predator-prey interactions in aquatic systems, and could be incorporated in a variety of biological and ecological studies on fish communities. For example, functional response mechanisms could be revealed from stomach content analysis in

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relation to prey abundance (Andersen, 1982), and feeding strategies of individual predators could be inferred from stomach content data (Temming & Mergardt, 2002). Such important ecological aspects are usually not considered a part of the motivation for the necessity of conducting detailed and apparently cumbersome laboratory experiments on gastric evacuation. The standard motivation for similar studies has been the need for an unbiased and precise estimation of food consumption rate.

An extensively applied approach to estimate food consumption rates of fishes in the wild is to combine information about amount and prey composition of their stomach contents with a gastric evacuation model (Bajkov, 1935; Elliott & Persson, 1978; Pennington, 1985). During the latest decades different gastric evacuation models based on substantial amounts of empirical information about evacuation in predatory gadoids have been put forward (Jones, 1974; Bromley, 1988; dos Santos & Jobling, 1995; Andersen, 2001).

A prerequisite for achievement of unbiased estimates of food rations is that the evacuation model is able to describe evacuation from the current amount of content in the stomach independently of (unknown) meal size (Pennington, 1985). Recent laboratory studies have demonstrated that evacuation rate in predatory gadoids described by the square root of current stomach content mass:

$$\frac{dS_t}{dt} = -\rho S_t^{0{\cdot}5} \tag{1}$$

possesses this ability (Andersen, 1998, 1999, 2001; Temming & Herrmann, 2003). S_t is total mass of stomach content at time t, and ρ is the rate parameter. This relationship together with most other evacuation models is, however, mainly based on empirical evidence. The exception is the surface-dependent evacuation model proposed by Salvanes *et al*. (1995). Incorporating prey geometry into the model, they regarded the stomach content as an ellipsoid body from which a layer of constant thickness δ was peeled off the surface per time unit as a result of the digestive processes in the stomach. The ability of the model to describe evacuation of a meal composed of one prey independently of prey size was evaluated by use of data from laboratory experiments performed by dos Santos & Jobling (1991) on cod *Gadus morhua* L. fed meals of one herring *Clupea harengus* L. They obtained wide confidence limits and concluded that δ could not be claimed to depend on prey size. Their estimates of δ, with the exception of the one obtained from the smallest herring, decreased, however, with increasing prey size. This was not the case with the square root model, which predicted similar values of the rate parameter for all prey sizes (Andersen, 1999). Furthermore, the square root principle excellently described situations where variations in meal size were a result of different numbers of prey (Andersen, 1998, 2001), and it seems therefore to have some generality.

Prey were regarded as homogenous units in all of the above considerations. Different prey characteristics are, however, known to influence gastric evacuation rate as summed up by Bromley (1994). The effect of prey energy density has been quantified by Andersen (1999, 2001) and Temming & Herrmann (2003), and the increased resistance to digestion of prey with robust exoskeletons has been documented in numerous studies. Energy density and resistance to digestion is not necessarily homogeneously distributed in the prey and the rate

parameter estimates described above represent average figures over total evacuation time.

In continuation of Salvanes *et al.* (1995) the objective of this study was to form the basis of a simple and conceptually consistent surface-dependent evacuation model with predictive power. It is argued that such a mechanistic gastric evacuation model can be represented by a simple geometric interpretation of the square root model considering total stomach content a cylinder. The model abstraction is that the curved side of the cylinder represents an outer surface of the stomach content that is equally exposed to the digestive processes. The possible influence of fish prey heterogeneity on the gastric evacuation pattern was examined as well, and it was suggested how the effects of prey characteristics might be incorporated into the square root model to produce a generic model of gastric evacuation.

MATERIALS AND METHODS

GASTRIC EVACUATION AND MEAL SIZE

The gastric evacuation data of dos Santos & Jobling (1991) used by Salvanes *et al.* (1995) to evaluate the performance of their ellipsoid evacuation model were obtained from cod of substantially different body masses and temperatures (Table I). No measurements were however taken by Salvanes *et al.* (1995) to account for these differences. In a first step to do so, power α of the general power model:

$$\frac{dS_t}{dt} = -\rho S_t^{\alpha} \tag{2}$$

was estimated from combined data on all herring sizes. S_t is total mass (g) of stomach content at time t (h), and $\rho(g^{1-\alpha}\ h^{-1})$ is the rate parameter to be estimated also. The estimations were performed using the rate parameter kept simple as well as expanded by $\rho = \rho_{MT} M^{0{\cdot}39} e^{0{\cdot}13T}$, where M and T are body mass (g) and temperature (° C) of cod, and ρ_{MT} is the mass- and temperature-specific rate parameter. The values of body mass power and temperature coefficient were those estimated by dos Santos & Jobling (1991) from a more comprehensive set of data including the data applied in the present context. Subsequently, rate parameter values of the ellipsoid model and the square root model were likewise estimated from data for each herring size as well as from the combined data for all sizes of herring to examine whether the estimates were independent of herring size as concluded by Salvanes *et al.* (1995) and Andersen (1999) for their respective models. In addition, the possible effects of herring size on the rate parameter estimates in the two models were quantified and tested expanding the parameters by a power function of herring size ($S_0{}^{\beta}$).

CONFIGURATION OF STOMACH CONTENT

In the present study, a minor experiment was performed on cod fed meals of either sandeel *Ammodytes tobianus* L. or herring to examine body positions of prey and characterize total stomach content. Herring and sandeel were sampled, frozen and stored as described by Andersen (1998, 1999) and Andersen & Beyer (2005). Cod were caught along the Danish Skagerrak coast by jigging with baited hooks and their husbandry was similar to that detailed by Andersen (1998, 1999). For the experiment, eight cod were kept together in each of four 2 × 2 m tanks at 10° C. After 4 days of food deprivation to ensure that no food from previous feedings was left in the stomach the cod were fed a number of sandeel or herring (Table II). Approximately 2 h subsequent to feeding each individual cod was anaesthetized (tricaine methanesulphonate, MS-222 Sandoz; dosage 1 : 16 000), total length (L_T) measured, and an incision was made along the *linea alba* starting from immediately anterior to the anus and stretching to a position between the

TABLE I. Basic experimental data and estimates of rate parameters δ and ρ in two gastric evacuation models to evacuation data on cod fed a meal of one herring

Herring		Cod		Ellipsoid model				Square root model				
Mass S_0 (g)	Total length l_0 (cm)	Mass M (range) (g)	Temperature T (range) (° C)	$\delta \times 10^{-3}$ (±95% CL) (cm h^{-1})	Adjusted r^2	$\delta_{MT}{}^* \times 10^{-4}$ (± 95% CL) (cm $g^{-0.39}$ h^{-1})	Adjusted r^2	$\rho \times 10^{-2}$ (± 95% CL) ($g^{0.50}$ h^{-1})	Adjusted r^2	$\rho_{MT}{}^* \times 10^{-3}$ (± 95% CL) ($g^{0.11}$ h^{-1})	Adjusted r^2	Observations (n)
4	9·3	698 (420–940)	3·5 (3·1–4·5)	5·6 (±0·9)	0·78	2·8 (±0·5)	0·72	5·0 (±0·8)	0·77	2·5 (±0·5)	0·71	17
8	11·4	691 (420–940)	5·2 (4·5–5·5)	6·1 (±0·9)	0·84	2·4 (±0·4)	0·81	6·1 (±0·9)	0·84	2·4 (±0·4)	0·81	21
16	14·0	709 (560–940)	4·3 (3·9–4·5)	4·9 (±0·5)	0·89	2·2 (±0·2)	0·91	5·4 (±0·6)	0·89	2·5 (±0·2)	0·91	20
32	17·3	746 (560–1042)	4·0 (3·5–4·2)	4·4 (±0·5)	0·89	2·0 (±0·2)	0·88	5·4 (±0·7)	0·89	2·4 (±0·3)	0·88	16
48	19·5	772 (570–1042)	5·5 (4·5–7·3)	4·6 (±0·5)	0·92	1·7 (±0·1)	0·94	6·0 (±0·6)	0·92	2·2 (±0·2)	0·94	21
4–48	9·3–19·5	723 (420–1042)	4·6 (3·1–7·3)	5·1 (±0·3)	0·94	2·2 (±0·1)	0·93	5·6 (±0·3)	0·95	2·4 (±0·1)	0·95	95

*δ and ρ were expanded as a function of predator (cod) mass M and temperature T by $\delta = \delta_{MT} M^{0.39} e^{0.13T}$ and $\rho = \rho_{MT} M^{0.39} e^{0.13T}$.

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TABLE II. Results from stomach configuration experiment on cod fed meals of sandeel or herring; figures in parentheses represent ranges

Prey				Cod			
Mass S_0 (g)	Length l_0 (cm)	Number offered in total	Number eaten per cod	L_T (cm)	Stomach length l_s (cm)	Stomach length 100 l_s L_T^{-1}(%)	Observations (n)
Prey: sandeel							
2·4	8·9	75	9	46·1	11·6	24·9	8
(2·3–2·5)	(8·8–9·2)		(1–13)	(42·4–48·9)	(10·6–12·5)	(23·4–26·0)	
7·7	12·9	28	4	47·9	11·9	24·5	8
(7·5–7·9)	(12·6–13·2)		(2–6)	(41·2–51·3)	(10·4–12·7)	(23·6–25·5)	
Prey: herring							
5·4	9·5	25	5	46·6	11·4	24·6	8
(5·2–5·5)	(9·4–9·7)		(2–8)	(43·5–50·2)	(10·7–12·6)	(23·1–25·8)	
15·0	13·2	15	2	48·6	12·2	25·2	8
(14·7–15·4)	(13·1–13·3)		(1–3)	(41·6–52·7)	(10·9–13·2)	(23·9–26·2)	

bases of the pelvic fins. The stomach shape was noticed and its length was measured from the sphincter between oesophagus and stomach to the posterior end of stomach. Finally, the stomach was opened by a longitudinal incision and prey number and configuration were registered. The fish was killed immediately thereafter.

MODEL DEVELOPMENT

The challenge is to interpret the main characteristics of gastric evacuation by a simple geometric abstraction of complex surface-dependent digestive processes. Ellipsoid geometry may appear as a promising candidate. It is, however, not obvious that this prey geometry is maintained during evacuation using a simple, spatially and temporally constant rate parameter δ. The dynamics of the rate parameter δ as applied by Salvanes *et al*. (1995) was therefore scrutinized. The outcome of this assessment together with the results from examination of stomach content configuration and the capabilities of the ellipsoid and square root models to describe evacuation independently of meal size were used to direct the model concept of surface-dependent evacuation towards a simple cylinder abstraction that described gastric evacuation in accordance with the square root model.

PREY HETEROGENEITY AND EVACUATION PATTERNS

The rate parameters of the ellipsoid and square root models represent mean evacuation rates over time of evacuation considering prey as homogenous units. For a variety of prey types, it has been shown that a constant value of the rate parameter of the square root model, this way, describes mean evacuation rate independently of meal and prey sizes (Andersen, 1998, 1999, 2001). This indicates that the basic pattern of evacuation in predatory gadoids may be described by the square root model. In reality, however, the rate parameter describing evacuation of individual meals most likely changes with time due to prey heterogeneity. Deviation from 0·5 of estimated power value as obtained by use of the general power model [equation (2)] to describe empirically the data on evacuation of an individual meal, thus, reflects prey heterogeneity that modulates the basic square root pattern. Values $<0{\cdot}5$ may for example be caused by delayed initial evacuation (exoskeletal barrier effect), whereas high values $>0{\cdot}5$ may be a result of high initial evacuation rate compared to the mean rate for the entire evacuation (*e.g.* initial evacuation dominated by materials of low energy such as entrails and ventral parts of the head region of juvenile fish prey).

Power estimates [α in equation (2)] were obtained from evacuation data (Table III) on whiting *Merlangius merlangus* (L.), cod and saithe *Pollachius virens* (L.) fed meals of fish prey (Andersen, 1998, 1999, 2001), and related to the prey body composition. The prey

TABLE III. Basic data on gastric evacuation experiments used to examine the effects of prey heterogeneity on evacuation pattern (see Andersen, 1998, 1999, 2001)

	Predator		Prey				
Temp. (° C)	L_T (cm)	Body mass (g)	Species	Item (*n*)	Body mass (g)	Energy density (kJ g^{-1})	Observations (*n*)
Predator : whiting							
15·0	34·5	468	Sandeel	2	4·50	5·63	26
12·5	34·8	482	–	2	4·51	5·63	23
12·0	26·6	163	–	2	1·85	4·38	32
9·8	35·0	491	–	2	4·51	5·63	25
9·0	27·4	181	–	2	1·85	4·38	33
9·8	35·4	513	–	1	4·49	5·63	28
9·4	35·9	547	–	2	1·85	4·38	24
6·2	29·1	228	–	2	1·84	4·38	24
9·6	29·2	231	–	2	4·50	4·99	21
9·6	33·8	385	–	2	4·50	4·99	29
12·2	32·8	365	–	2	2·50	4·62	16
12·5	25·4	152	–	2	2·01	4·64	18
12·3	37·8	560	Sprat	1	14·0	11·5	18
12·8	37·4	602	Herring	2	4·51	4·08	24
9·7	35·5	445	–	2	4·50	4·69	38
9·8	31·8	348	–	1	4·01	3·71	22
9·7	32·4	375	–	2	4·00	3·71	27
9·6	37·1	606	Norway pout	2	5·16	5·04	33
12·2	30·7	297	–	2	2·60	3·80	15
12·2	31·1	315	–	1	5·17	4·08	15
9·8	36·3	492	Whiting	2	4·51	3·44	33
12·5	23·8	118	Brown shrimp	5	0·50	4·08	25
12·5	31·7	293	–	5	0·50	4·08	25
12·5	36·8	512	–	5	0·50	4·08	28
12·5	33·2	412	–	2	0·50	4·08	28
12·5	33·3	416	–	10	0·50	4·08	25
12·2	31·4	274	–	6	0·75	3·97	22
12·2	30·9	269	–	2	0·75	3·97	30
Predator : cod							
12·3	28·9	233	Sandeel	2	4·51	6·60	21
10·2	49·4	1234	–	2	8·01	5·98	16
12·3	47·1	1081	Sprat	2	14·0	11·5	22
12·3	30·0	265	–	1	11·1	10·6	17
12·3	29·7	256	Sprat+herring	2	10·6	8·00	16
12·3	30·9	304	Herring	2	9·98	5·10	15
12·3	28·4	217	–	2	4·49	4·00	20
Predator : saithe							
10·0	41·3	683	Norway pout	2	4·00	4·30	23
10·0	42·8	712	–	1	4·00	4·30	26

species were sandeel, herring, sprat *Sprattus sprattus* (L.), Norway pout *Trisopterus esmarki* (Nilsson) and whiting. Data on evacuation of meals of one or two fish prey items, exclusively, were used here, because higher numbers of prey in a meal tended to give power estimates closer to 0·5 probably because temporally differentiated evacuation of body parts between prey blurred the possible effect of their heterogeneity. In addition, power estimates were obtained from data on evacuation of meals of brown shrimp *Crangon crangon* (Fabricius) in whiting (Andersen, 1998, 1999, 2001) to illustrate the effect on evacuation pattern of a robust exoskeleton. Body composition of the fish prey was examined in the present study. The prey used for this purpose were sampled from the same sea areas at the same times of the year (month) as those of the evacuation studies. They were frozen immediately after sampling and stored at −30° C. Then, the prey were thawed in water, and 30–70 specimens of a species, covering the range of body sizes used in the evacuation experiments, were selected and individually weighed after dabbing with a moist paper towel. Head, internal body organs (entrails), tail fin, and pectoral girdle and fins were dissected from the trunk. The difference between masses of total body and trunk fraction was considered the mass of the 'rest' section assuming that possible dissection loss was mainly body fluids originating from the rest of the fraction. Each of the two fractions from 10 specimens forming a section of the applied body size range were grouped and dried at 60° C to constant mass. Then, energy density was determined by bomb calorimetry in an IKA C-7000 bomb calorimeter.

STATISTICAL ANALYSES

Parameters of the general power evacuation model were estimated by non-linear regression using the integrated version of equation (2):

$$S_t = S_0\left[1-(1-\alpha)\rho t {S_0}^{\alpha-1}\right]^{(1-\alpha)^{-1}}+\varepsilon \tag{3}$$

where S_t is mass of total stomach content at time t after ingestion of a meal of mass S_0, and ϵ is considered normally distributed with a mean of zero and a S.D. that is estimated by the regression residuals. The parameters of the ellipsoid model and the square root version of the general power model were estimated likewise using equations (5) and (10) (see Results). According to Andersen (1998), the variation of S_t increases in general proportionally to S_0 (*i.e.* $\epsilon \propto S_0$). S_t was therefore expressed relative to S_0 to obtain homogeneity of variances whenever evacuation data on different meal sizes entered the analyses. The data sets from each of the experiments were truncated at the time of first occurrence of an empty stomach (Andersen, 1984; Olson & Mullen, 1986; Bromley, 1994). The model parameters were estimated using the iterative Marquardt method in the NLIN procedure of SAS (1990) with the modifications described in Andersen (1998). The non-linear coefficient of determination r^2 was adjusted to account for the number of estimated parameters according to Sokal & Rohlf (1981).

RESULTS

GASTRIC EVACUATION AND MEAL SIZE

The estimated value of the power α using equation (3) to combined gastric evacuation data on all sizes of herring was 0·55 (Table IV). Correcting for the effects of size and temperature of cod an estimate of 0·49 was obtained. This means that the rate parameters of power models with a power close to 0·5 provide the best description of evacuation of the herring meals independently of prey and meal sizes. The estimates of the rate parameter ρ in the square root model as obtained from data on each of the size groups of herring were therefore

TABLE IV. Summary statistics for parameter estimation of the power model to combined evacuation data for all meal sizes on cod fed a meal of one herring

Model version	α (±95% CL)	ρ or $\rho_{MT} \times 10^{-3}$ (±95% CL)	Adjusted r^2	Observations (n)
Simple ρ	0·55 (±0·06)	50 (±7)	0·95	95
ρ_{MT}*	0·49 (±0·06)	2·5 (±0·3)	0·95	95

*ρ was expanded as a function of predator mass M and temperature T by $\rho = \rho_{MT} M^{0\cdot39} e^{0\cdot13T}$.

similar (Table I). In consequence, the power β quantifying the effect of meal size on ρ was not significantly different from 0 using the square root model to combined data on all herring sizes (Table V). Hence, total evacuation time τ could be regarded as being proportional to the square root of herring size [$\tau = 2S_0^{0\cdot5}\rho^{-1}$ as derived from equation (10)].

CONFIGURATION OF STOMACH CONTENT

The number of prey allotted to each tank was limited to ensure large variation in amount of ingested food among cod (Table II). The posterior end of stomach was invariably located just anterior to the anus independently of type, size and number of prey, and stomach length made up *c.* 25% of the L_T of cod (Table II). The stomach contracted relatively quickly after incision into the prey. It was, however, in general possible to observe position and geometry of the prey. All prey were more or less folded back with the bends turning posteriorly in the cod and the prey ends were oriented in the opposite direction along the length axis of the stomach. Smaller meals tended to accumulate in the posterior part of the stomach leaving free space in the anterior part, and even prey of single-prey meals were folded. The outer shape of the stomach was more or less irregular reflecting the actual configuration of prey inside the stomach (Fig. 1).

TABLE V. Estimates of the effect of meal size S_0 on rate parameters δ and ρ as expressed by $\delta = \delta_{S_0} S_0^{\beta}$ and $\rho = \rho_{S_0} S_0^{\beta}$ in two gastric evacuation models to combined evacuation data for all meal sizes on cod fed a meal of one herring

Model version	δ_{S_0} or δ_{MTS_0}* $\times 10^{-4}$ (±95% CL)	ρ_{S_0} or ρ_{MTS_0}* $\times 10^{-3}$ (±95% CL)	β (±95% CL)	Adjusted r^2	Observations (n)
Ellipsoid model					
δ_{S_0}	69 (±12)		−0·12 (±0·06)	0·95	95
δ_{MTS_0}*	3·7 (±0·7)		−0·19 (±0·06)	0·95	95
Square root model					
ρ_{S_0}		51 (±9)	0·03 (±0·06)	0·95	95
ρ_{MTS_0}*		2·7 (±0·5)	−0·04 (±0·06)	0·95	95

*δ_{S_0} and ρ_{S_0} were expanded as a function of predator mass M and temperature T by $\delta_{S_0} = \delta_{MTS_0} M^{0\cdot39} e^{0\cdot13T}$ and $\rho_{S_0} = \rho_{MTS_0} M^{0\cdot39} e^{0\cdot13T}$.

(a)



(b)



FIG. 1. Ventral view of stomach of 45 cm total length cod fed a meal of 9 cm sandeel 2 h prior to photography. (a) Stomach shape and (b) prey configuration inside opened stomach are indicated.

LESSONS LEARNED FROM THE ELLIPSOID MODEL

Salvanes *et al.* (1995) expressed the initial mass S_0 (g) of a fish prey by the volume of an ellipsoid of revolution

$$S_0 = 1{\cdot}33\pi r_0^2 0{\cdot}5 l_0 \tag{4}$$

considering the prey to be homogenous, of ellipsoid shape, and of a mass density of 1 (g cm^{-3}). Initial L_T (cm) of prey was l_0, and r_0 indicated radius (cm) estimated from an ellipsoid of mass S_0 and length l_0 along axis of rotation. Then, tacitly assuming ellipsoid geometry throughout evacuation, mass S_t (g) of remaining prey in the stomach at time t was predicted to be

$$S_t = 1{\cdot}33\pi(r_0 - \delta t)^2(0{\cdot}5l_0 - \delta t) \tag{5}$$

where δ was the evacuation rate parameter (cm h^{-1}). Apparently, the ellipsoid model seems to be simple and attractive to use. As shown in Appendix, the value of δ varies, however, in space to maintain an ellipsoid shape of the stomach content throughout evacuation. This means that the ellipsoid model in equation (5) is valid only if δ strictly refers to ends and middle of the ellipsoid with a lower evacuation rate elsewhere. The minimum δ-value decreases and its location shifts towards the ends of the ellipsoid with increasing slenderness of the ellipsoid. Application of the model implies, in consequence, complex spatial and temporal model patterns of surface-dependent gastric evacuation, which obviously was not the intention by Salvanes *et al.* (1995).

The second point is that the evacuation rate parameter δ of the ellipsoid model must decrease with increasing meal size S_0 to explain an evacuation time τ that is closely proportional to the square root of meal size. To arrive at this, note that in the ellipsoid model

$$\tau = r_0\delta^{-1} \tag{6}$$

because it follows from equation (5) that $S_t = 0$ for $r_0 - \delta t = 0$ as $l_0 \geq 2r_0$ for herring. Rearrangement of equation (4) give

$$r_0 = [1.5\pi^{-1}]^{0{\cdot}5}(S_0 l_0^{-1})^{0{\cdot}5} \tag{7}$$

Combining equations (6) and (7), it appears that evacuation time $\tau \propto S_0^{0{\cdot}5} {l_0}^{-0{\cdot}5}\delta^{-1}$. Length-mass relationship of fishes are usually expressed by a power function as $S_0 = al_0^b$. Thus, $\tau \propto S_0^{0{\cdot}5}S_0^{\beta}\delta^{-1}$, where $\beta = -(2b)^{-1}$. In consequence, δ must be proportional to S_0^{β} to make τ proportional to $S_0^{0{\cdot}5}$. Salvanes *et al.* (1995) used the value 3·358 of power b for herring prey. In this case, δ may be expected to be proportional to $S_0^{-0{\cdot}15}$ assuming that τ in reality is proportional to the square root of S_0. The effect of herring size S_0 on δ was significant using the ellipsoid model to combined data on all herring sizes, and the estimated value of β was not significantly different from −0·15 (Table V). This applied whether or not corrections were made for variations in size and temperature of cod. The estimates of δ decreased accordingly with increasing size of herring (Table I) when corrected for the effects of differences in body mass and temperature of cod by estimating δ_{MT} (Table I).

The results from the prey configuration trials of the present study showed that maximum length of the stomach amounted to *c.* 25% of L_T of cod and that the prey tended to fold up inside the stomach. The cod applied by dos Santos & Jobling (1991) in their evacuation experiments reported here ranged from 40 to 50 cm L_T (Salvanes *et al.*, 1995). Maximum length of the stomachs was then 10–12 cm, which means that at least all herring belonging to the three larger size groups (Table I) have been folded up to some extent, and that the initial stomach

content lengths after ingesting a herring were similar irrespective of herring size. A constant initial length l_0 of stomach content independent of the size of the ingested herring would actually imply that the ellipsoid model describes evacuation time to be proportional to the square root of herring size ($\tau \propto S_0^{0{\cdot}5} l_0^{-0{\cdot}5} \delta^{-1}$ as shown above but with constant l_0) in accordance with the empirical findings. It also means, however, that the ellipsoid seems excessive as a general descriptor of more or less folded up prey in the stomach forming an irregular geometry of total stomach content. Together with the unsubstantiated complex spatial and temporal variation in the values of the rate parameter of the ellipsoid model (Appendix), this leads to the search for a more simple geometric interpretation of the square root model.

NEW MODEL

A simple abstraction of the square root model is to regard the stomach content as a homogeneous cylinder of 1 g cm^{-3} the mass of which is gradually reduced by successive peeling off its curved side whereas the ends are left unaffected. The mass S_t of a cylinder of length l and radius r_t is $\pi r_t^2 l$, and the surface area A_t of its side is $2\pi r_t l$. The instantaneous evacuation rate (g h^{-1}) may then be described by the cylinder model as

$$\frac{dS_t}{dt} = -\rho {S_t}^{0{\cdot}5} = -\delta 2\pi r_t l = -\delta A_t \tag{8a}$$

where

$$\delta = \rho(4\pi l)^{-0{\cdot}5} \tag{8b}$$

Here, the rate parameter δ (cm h^{-1}) denotes the depth of the layer that is peeled off the cylinder side per time unit. The integrated version over time is

$$S_t = \pi(r_0 - \delta t)^2 l = \pi {r_0}^2 l\left(1 - \delta t {r_0}^{-1}\right)^2 \tag{9}$$

and evacuation time is $\tau = r_0 \delta^{-1}$. Notice that δ is inversely proportional to the square root of l [equation (8b)] and that the value of ρ is independent of the proportions of the cylinder. This is a consequence of the general ability of the square root model to describe evacuation independently of prey configuration as shown for a variety of prey species fed to different gadoids. Combining equations (8b) and (9), the surface-based cylinder model may be expressed directly by the mass-based square root model:

$$S_t = S_0\left[1 - \rho t (2S_0^{0{\cdot}5})^{-1}\right]^2 \tag{10}$$

which is equivalent to integrated version of equation (1).

The cylinder model would also apply if the cylinder ends were digested at the same rate as the sides. This model version would comply with the empirical relationship between the square root of meal size and evacuation time if $\delta \propto l^{0{\cdot}5}(l + r)^{-1}$, which may be deduced like it was done for the cylinder model with ends unaffected [equation (8)]. The latter model formulation was,

however, put forward here partly because observations from previous gastric evacuation experiments strongly indicate that it is primarily the sides of the prey that are digested (unpubl. obs.). This may be a result of the prey ends being oriented along the length axis of the stomach and thus escaping direct contact with the stomach wall. As discussed below, the other reason for this choice of model version is that the proportions of the cylinder representing total stomach content, in principle, do not matter because the stomach length is constant irrespective of the amount of stomach content.

PREY HETEROGENEITY AND EVACUATION PATTERNS

Variances of the power estimates obtained from evacuation data on fish prey were not significantly different (Bartlett's test; $f = 29$; $P > 5\%$) and common S.E. (0·093) as well as 95% CL (±0·19) could be estimated putting n at the mean number of observations (23) of an experiment. Power values estimated from small sandeel ranging up to 2 g were situated well above 0·5, whereas sandeel of 4·5 g or larger gave values of c. 0·5 (Fig. 2). In contrast, high values were generally obtained from herring, Norway pout and whiting of 4–5 g. The exception was the estimate of 0·44 obtained from one of the experiments on 5 g Norway pout. The energy density of this batch of Norway pout was high (5·04 kJ g^{-1}) as compared to the other batches of Norway pout as well as to whiting (Table 1). Larger herring and sprat gave power values of c. 0·5. This trend was confirmed by the herring data of dos Santos & Jobling (1991) that were used in the model

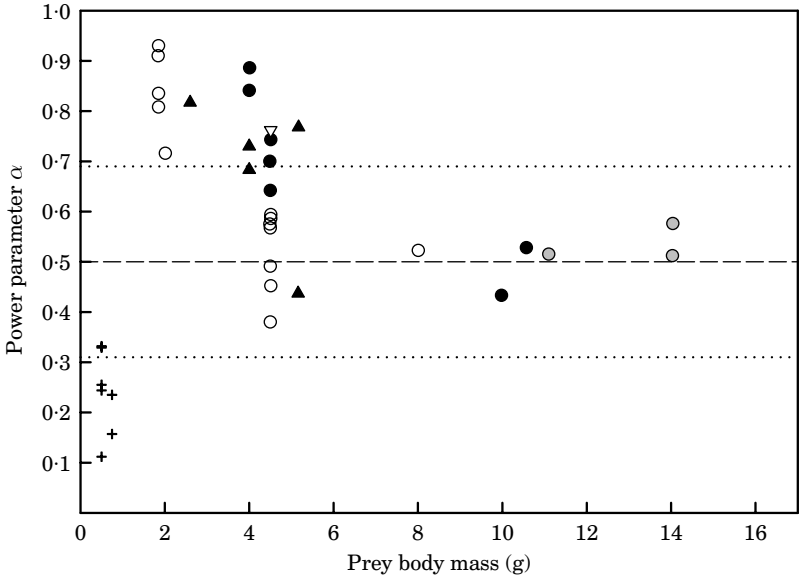


FIG. 2. Relationship between prey body mass and estimated power value obtained by application of a power model to data on gastric evacuation in predatory gadoids fed meals of sandeel (○), sprat (◍), herring (●), Norway pout (▲), whiting (▽) and brown shrimp (+). Common 95% CI (·····) for all power estimates is placed around the power 0·5 (– –) of the square root version of the power model to indicate the significance of deviation of individual power estimates from this power.

development. Herring of 4, 8, 16, 32 and 48 g gave power estimates of 1·16, 0·75, 0·54, 0·47 and 0·65, respectively. Because of large data scatter, the 95% CI of these estimates were, however, wide with lower limits of −0·31–0·32 and upper limits lying between 0·98 and 2·02. The estimates obtained from brown shrimp were generally low ranging from 0·11 to 0·33 (Fig. 2).

The trunk part of total body mass increased with increasing body size for all prey fishes (Fig. 3). In contrast to the patterns obtained from sprat and herring initial increase in trunk fraction for sandeel was very fast attaining a level >80% at a body mass of 4 g. For herring and sprat a body size of *c.* 10 g was required to obtain shares of 75–80% of total body mass. The trunk fractions of the two gadoid prey were lower than those of the other prey. The fraction increased, however, at a faster rate for Norway pout than it did for whiting.

Energy density of total body also increased with increasing body size for all prey fishes (Fig. 4), although it was much more pronounced for the small-sized species sandeel, sprat, and (to a lesser extent) Norway pout than it was for the medium-sized fishes whiting and herring. Energy densities of the trunks of gadoids and herring were *c.* 4 kJ g^{-1} for all body sizes, while they increased substantially with increasing body sizes for sandeel and, especially, sprat. The energy density of the 'rest' fraction (head, entrails, tailfin, and pectoral girdle and fins) was generally lower than that of the trunk except for sandeel and Norway pout, where this relationship between the two fractions reversed at body masses of *c.* 2–8 g, respectively.

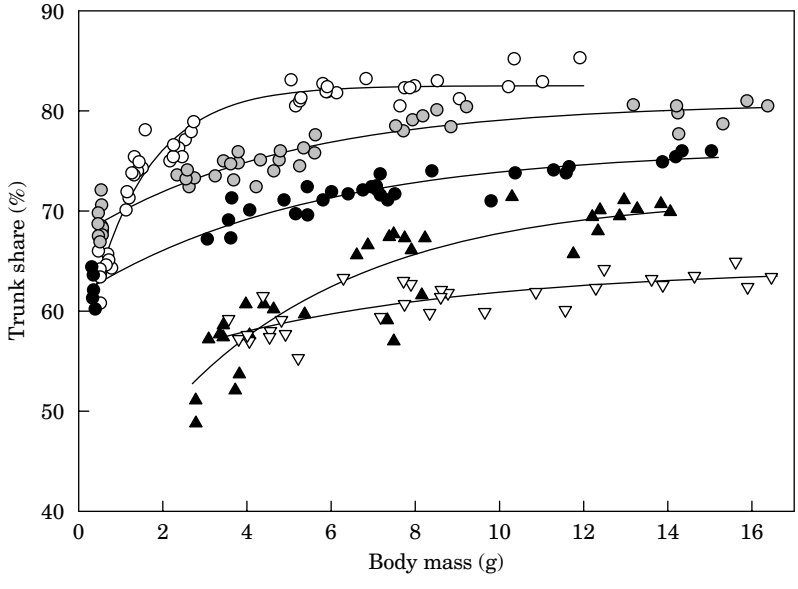


FIG. 3. Per cent trunk part of total body mass depicted as a function of total body mass for sandeel (○), sprat (◯), herring (●), Norway pout (▲) and whiting (▽). Trunk is total body minus head, internal body organs, tail fin, and pectoral girdle and fins. Curves are provided using the expression $P = a[1 - be^{(cM-d)}]$ to data for each prey species, where M and P are total body mass and percentage of trunk, and the other symbols denote estimated parameters.

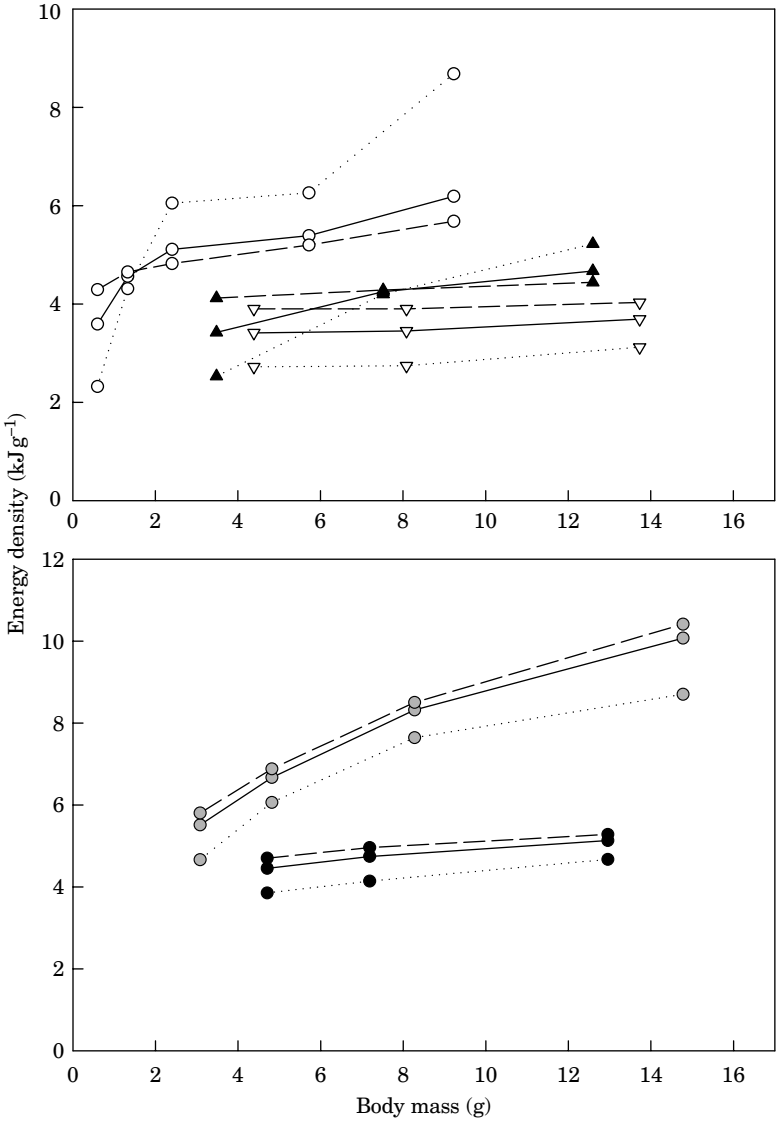


FIG. 4. Energy densities of total body and body parts depicted as a function of total body mass for sandeel (○), sprat (◯), herring (●), Norway pout (▲) and whiting (▽). Trunk part (– –) is total body (—) minus head, entrails, tail fin, and pectoral girdle and fins (····).

DISCUSSION

This study provides a mechanistic model approach to gastric evacuation in predatory fishes that is based on a simple geometric abstraction of complex surface-dependent digestive processes, complying with the square root principle. Digestive processes are assumed to be restricted to an outer surface of total stomach content that is represented by the curved side of a cylinder. This way, total stomach content is considered a cylinder the mass of which is gradually

reduced by successive peeling off its side leaving the ends unaffected. The spatially and temporally constant rate parameter δ describes the thickness of the layer that is peeled off the cylinder per time unit. The present results demonstrated that the ellipsoid prey geometry of Salvanes *et al.* (1995) is rather intricate in that it is not maintained during evacuation unless the fundamental rate parameter varies significantly in space in a complex way that has not been substantiated.

The square root model describes appropriately gastric evacuation independently of meal size in different gadoids for a variety of prey species predicting that total evacuation time is proportional to the square root of meal size (Jones, 1974; Andersen, 1998, 1999, 2001; Temming & Herrmann, 2003). The present study showed that this relationship applied also to the evacuation data of dos Santos & Jobling (1991) on cod fed herring that were used by Salvanes *et al.* (1995) in their attempt to validate their ellipsoid model. It is possible to formulate an ellipsoid model that satisfies the square root principle assuming constant length of the ellipsoid. This involves a δ-dynamics where the value of the fundamental rate parameter increases monotonically from zero at the ends to a maximum value at the middle of the ellipsoid (may be deduced as it was done for the ellipsoid model version in Appendix). The prey configuration trials, however, did not support the ellipsoid geometry of total stomach content as a general feature. The results showed an irregular geometry reflecting more or less folded and packed prey in the stomach. It seems therefore useless to search for an invariant geometry that fits the shape of stomach content throughout evacuation. In consequence, a cylinder abstraction was put forward here.

An inherent characteristic of the cylinder interpretation of the square root model is that the rate parameter δ is inversely proportional to the square root of the length of the stomach content. It is simply the consequence of the general ability of the square root model to describe evacuation in predatory gadoids independently of prey configuration. Anatomical and dynamic characteristics of the stomach render this formulation of δ plausible. The stomachs of predatory gadoids included in this study are J-shaped, and the absolutely dominating *pars cardiaca* of these stomachs forms a tube in which the mucosa is thrown into longitudinal folds. The anatomical organization corresponds well with the present finding on cod that the tube length seems to be constant and independent of the amount of content, whereas the mean radius of the tube in consequence increases with increasing amounts of content. Thus, unfolding of the longitudinal folds of mucosa probably increases the mucosal surface in contact with the stomach content concurrently with the increasing surface of the content, which supports the model prediction of constancy of the rate parameter for different meal sizes of similar lengths and throughout evacuation of a meal as long as its length remains constant. The above applies for large contents that fill the entire length of the tube. For small contents that occupy the posterior part of the stomach only, the length l of the content is consequently smaller than that of the stomach. According to the cylinder model the rate parameter δ of such a content should be higher than that of a large content packing the entire stomach because $\delta \propto l^{-0{\cdot}5}$. The relationship might be due to these smaller contents being shuttled inside the stomach during the peristaltic movements increasing the chemical, enzymatic and mechanical contact between mucosa of the stomach wall and the

content. The increase in efficiency of the digestion processes might in this way fully offset the proportions of the content that do not allow it to fill the entire length of the stomach. Although this dynamic of stomach content is not formally proven, it complies with the extensive empirical evidence of the relationship between square root of meal size and evacuation time for a variety of meal sizes and prey types. Then, in principle, length of the stomach content and proportions of the cylinder representing the content do not matter according to the cylinder model.

Estimates of total food rations might be obtained using the mass-based square root equation (1) directly to data on total stomach content. A geometric interpretation is, however, necessary if evacuation of individual prey from the stomach are to be quantified, *i.e.* size and proportions of each prey should be used to estimate its contribution to the surface area exposed to digestive processes. With ends kept unaffected in the model individual prey may be allowed to break into smaller pieces without the need to change the value of the rate parameter δ, because the extra end areas introduced by the fragmentation do not matter, and the prey may thus be regarded as one complete item throughout evacuation. Being a geometric interpretation of the square root principle the capability of this simple cylinder model cannot be challenged in the present context. Andersen & Beyer (2005), however, applied the cylinder model to predict evacuation of individual prey in mixed meals of different prey types.

In all of the above model considerations possible heterogeneities of prey that may influence the instantaneous rate of evacuation were disregarded using mean evacuation rate over evacuation of a meal. The present study confirmed that evacuation patterns vary between prey types, and the results indicated that the variations were related to characteristics of different body parts as suggested in the literature (Bromley, 1994; Salvanes *et al.*, 1995). High trunk share of total body mass in large prey fishes is in accordance with the fact that percentage of muscle increases with increasing body size (Brett & Groves, 1979). In contrast to the other body parts that are composed of a variety of tissue types, body muscle mass is probably forming a homogenous unit regarding digestibility. The very high trunk share of body mass in 4·5 g sandeel may, therefore, explain the estimated power values of *c.* 0·5 (square root relationship) as opposed to the values obtained from herring, Norway pout and whiting of 4–5 g. At this body size, a substantial percentage of body mass of the three latter species was made up by head, fins and entrails, which in general were digested at an early time during the process of evacuation as opposed to the compact body muscles. This may explain the higher power values obtained from these more heterogeneous prey. The deviating, low value of 0·44 obtained from the batch of high energy Norway pout of 5 g was probably a result of initial evacuation rate being delayed by digested liver material of high energy density. Energy density of muscle tissue in gadoids is quite constant over body size (and close to 4 kJ g $^{-1}$) as indicated in this study also, while accumulated lipid in the liver acts as the energy reserve. The trunk shares in larger clupeids were high, and power estimates obtained from these prey, as opposed to results from small herring, were accordingly centred on the square root value of 0·5. High heterogeneity of small fishes in terms of small trunk share of body mass may explain why Ruggerone (1989) and Elliott (1991) achieved exponential evacuation for

salmonids foraging on fish prey of <1 g. Jobling (1987) suggested that curvature of the evacuation curve decreases with increasing size of prey, and that evacuation of large prey may be described appropriately by a more linear track. Livers of high energy density delaying initial evacuation of otherwise homogeneous prey fishes might explain that a more linear evacuation pattern was suitable for larger gadoids ($\alpha < 0{\cdot}5$), whereas the present results do not point to power values <0·5 for larger clupeids, because their trunks contain more energy than the rest of their bodies. Resistant exoskeletons may modify the underlying trend in gastric evacuation in the same way as high energy livers (Bromley, 1994). Evacuation of shrimps and crabs have therefore been described by a linear model or by use of a power exponential predicting a sigmoidal pattern, which indicates that a substantial delay may take place during the first part of evacuation (Hopkins & Larson, 1990; dos Santos & Jobling, 1992). In the present study, low values of the power parameter was obtained from data on evacuation of brown shrimp indicating that some initial delay modified a basic square root pattern.

Application of different power models to describe evacuation patterns of different heterogeneous prey types is, however, mainly an empirical approach. Using the square root model as the starting point for evacuation of homogeneous meals, a mechanistic approach to the formulation of a generic evacuation model may be to modulate the rate parameter as a function of prey characteristics that influence evacuation. The extended cylinder interpretation of the square root model presented by Andersen & Beyer (2005) to describe gastric evacuation of mixed contents in predatory gadoids included in this way the effects of body size, energy density and resistance to digestion of individual prey. The individual prey were considered homogeneous. The model could in principle be used to describe evacuation patterns of individual heterogeneous prey as well. This would, however, demand that the geometry of different body compartments differing substantially by their characteristics were taken into consideration in the model formulation. To utilize the potential of the model to estimate food rations detailed information about body composition of the prey organisms would be needed also.

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Appendix

Spatial and temporal variations in rate parameter δ of the ellipsoid model.

An ellipsoid is a solid of revolution. It is therefore sufficient to examine the ellipse represented by a longitudinal section of the ellipsoid in the median plane along the rotation axis in Fig. A1. Because of symmetry it is adequate to consider the first quadrant of the co-ordinate system only ($0 \leq \theta \leq \pi/2$). Thus, a two-dimensional outline of an ellipsoid-shaped meal at time 0 is described by the equation of an ellipse as:

$$x_0^2 a_0^{-2} + y_0^2 r_0^{-2} = 1 \tag{A1}$$

where initial meal length l_0 is $2a_0$ and radius is r_0. In order to maintain the ellipsoid shape throughout evacuation the points (x_t, y_t) on the surface of the stomach content need to lie on the ellipse at any time t:

$$x_t^2 (a_0 - \delta t)^{-2} + y_t^2 (r_0 - \delta t)^{-2} = x_t^2 a_t^{-2} + y_t^2 r_t^{-2} = 1 \tag{A2}$$

where δ is the rate parameter as defined by Salvanes *et al.* (1995). Consequently, considering the components $(\delta_{x,\theta}, \delta_{y,\theta})$ of the local rate δ_θ (cf. Fig. A1), the point $(x_{t+\Delta t}, y_{t+\Delta t}) = (x_t - \delta_{x,\theta}\, \Delta t,\ y_t - \delta_{y,\theta}\, \Delta t)$ must for small Δt lie on the ellipse specified by $(a_{t+\Delta t},\ r_{t+\Delta t}) = (a_t - \delta \Delta t,\ y_t - \delta \Delta t)$. Inserting this point in the equation for the ellipse at time $t + \Delta t$, Taylor developing to the first order (or differentiating with respect to t), reducing and utilizing equation (A2), it follows that $(\delta_{x,\theta}, \delta_{y,\theta})$ and δ must satisfy the relationship:

$$x_t \delta_{x,\theta}\, a_t^{-2} + y_t \delta_{y,\theta}\, r_t^{-2} = \delta\left(x_t^2\, a_t^{-3} + y_t^2\, r_t^{-3}\right)$$

Utilizing $\delta_{y,\theta} \delta_{x,\theta}^{-1} = y a^2 (x r^2)^{-1}$, it follows that the left-hand side is proportional to $(\delta_{x,\theta}^2 + \delta_{y,\theta}^2)^{0.5} = \delta_\theta$,

$$x_t \delta_{x,\theta} a_t^{-2} + y_t \delta_{y,\theta} r_t^{-2} = \delta_\theta \left(x_t^2 a_t^{-4} + y_t^2 r_t^{-4}\right)^{0.5}$$

This shows that the requirement in equation (A2) is fulfilled only when

$$\delta_\theta = \delta[(\cos^2\theta) a_t^{-1} + (\sin^2\theta) r_t^{-1}][(\cos^2\theta) a_t^{-2} + (\sin^2\theta) r_t^{-2}]^{-0.5} \tag{A3}$$

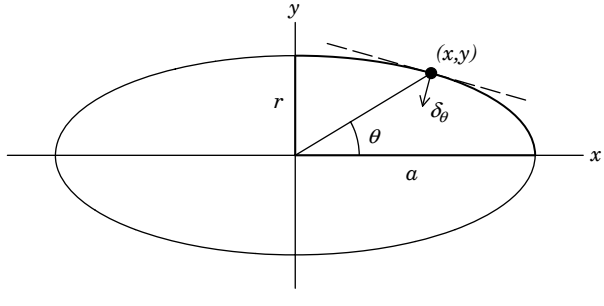


Fig. A1. Two-dimensional presentation of ellipsoid-shaped stomach content of length $l = 2a$ and radius r. The δ_θ -arrow indicates the local rate parameter, perpendicular to the tangent (––) at point (x, y) on the ellipse positioned by angle θ.

where δ_θ is the local rate parameter at time t at point $(x_t, y_t) = (a_t \cos\theta, r_t \sin\theta)$. It follows from equation (A3) first that $\delta_0 = \delta_{\pi/2} = \delta$ but δ_θ otherwise decreases with time and secondly for any point in time that δ_θ attains a minimum value of

$$\delta_{\min} = \delta 2(r_t a_t)^{0{\cdot}5}(r_t + a_t)^{-1} \tag{A4}$$

at

$$\theta_{\min} = \sin^{-1}\left[r_t(r_t + a_t)^{-1}\right]^{0{\cdot}5} \tag{A5}$$

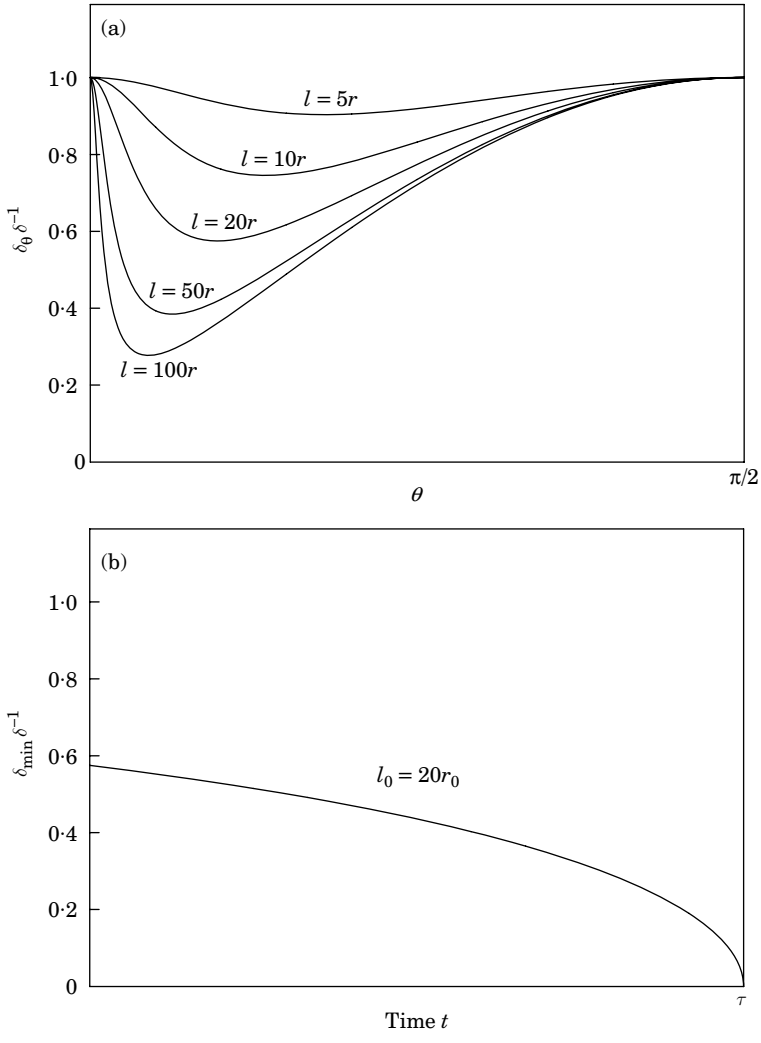


FIG. A2. (a) Spatial variation in local rate δ_θ (cm h^{-1}) at which surface layer is peeled off stomach content of length l and radius r during gastric evacuation according to surface-dependent ellipsoid evacuation model. δ_θ is expressed relative to rate at ends ($\theta = 0$) and middle ($\theta = \pi/2$) of the ellipsoid-shaped stomach content, which is the rate δ defined by Salvanes *et al.* (1995). (b) Temporal development of spatial minimum rate $\delta_{\min}$ expressed relative to δ during evacuation of a meal of an initial $l_0 : r_0$ ratio of 20. Evacuation time is $\tau = r_0\delta^{-1}$.

From this it appears that $\delta_{\min} = \delta$ if $a = r$ (sphere geometry of prey), and that $\delta_{\min}\delta^{-1}$ decreases with increasing value of $a\ r^{-1}$ (increasing slenderness of prey). Fig. A2(a) demonstrates the spatial variation of $\delta_\theta\ \delta^{-1}$ for a range of $l : r$ ratios of stomach contents ($l = 2a$). The initial $l_0 : r_0$ ratio of herring prey used by dos Santos & Jobling (1991) was *c*. 20. The ellipsoid evacuation model predicts increasing slenderness with time. The associated temporal development of $\delta_{\min}\ \delta^{-1}$ during evacuation of a meal of a single herring is shown in Fig. A2(b).